Certified Correction of Vision–Language Hallucinations: Gains, Preconditions, and a Sequential Failure Mode

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Abstract

Selective prediction equips a vision–language model (VLM) with a distribution-free guarantee on the answers it emits, and pays for that guarantee by abstaining on the rest. A recent impossibility result shows the bill cannot be reduced by better engineering: when the base risk μ exceeds the target α, *any* predictor that may only *emit or abstain the model's own answer* must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs, whatever its score, bound, or calibration budget. This paper takes that premise seriously and asks what changes when the predictor is permitted to emit a *different* answer. We introduce Certified-Correction Risk Control (CCRC): accept the model's answer where a correctness score is high, substitute an answer supplied by an *independent* evidence channel where that channel's evidence is strong, abstain otherwise, and certify the error rate of everything emitted at level α with confidence $1 - \delta$.

We prove validity for CCRC, characterise its behaviour, and evaluate it across five benchmarks, two backbones and a mitigation decoder. Four findings organise the paper. First, the correctness *score* matters more than the procedure: a detection-grounded score improves certified coverage over model-internal uncertainty by 10.8–32.1 points, and we decompose our total margin over a published CLIP-based scorer to show that 62.6 of 67 points come from the score rather than from our algorithm. Second, repairs cannot be self-certified — we prove that certifying a substitution with the complement of the score that flagged it is circular, and confirm that the resulting procedure repairs nothing. Third, where CCRC helps it does so by *dilution*: 0.9–1.8% of items repaired at a strict evidence gate buys 1.9–3.4× that mass in certified coverage, because high-precision repairs lower the emitted set's average risk and relax the binding constraint; we give a diagnostic, checkable before calibration, that is consistent with both the gains and the single dataset on which CCRC loses. Fourth, the natural generative extension fails: in a paired matched-prefix experiment, repairing a claim mid-caption significantly *increases* downstream hallucination ($+0.207 \pm 0.091$ mentions, $n = 121$, $p = 0.025$), which closes off a fixed-point calibration argument (correction-map monotonicity, not the risk monotonicity in λ that our one-shot guarantee avoids needing) and exposes an unpriced cost that a context-blind sequential procedure must charge for. We also reproduce the closest concurrent method and find the two mechanisms are complementary rather than competing. Code, extracted per-item signals, and all negative results are released.

1 Introduction

1.1 A small failure, and three unsatisfying options

Image 518 of the POPE benchmark is an office desk. On it, in plain view and occupying a respectable fraction of the frame, is a computer mouse. We ask LLaVA-1.5-7B: *"Is there a mouse in the image?"* The model answers **no**, and it is not especially confident about it — the probability mass on `yes` is 0.207. Our calibrated correctness score for this item is $s = 0.18$, in the bottom fifth of the distribution. An open-vocabulary detector, asked independently whether it can localise "a photo of a mouse" in the same pixels, returns a confidence of 0.903 with a margin of 0.753 over its own negative-class baseline. The object is there. The detector sees it. The model does not.

Now suppose we must ship this system with a promise: *among the answers we actually emit, at most* 10% *are wrong, with* 90% *confidence, and this must hold for the next batch of inputs rather than the one we tuned on.* Such promises are what conformal risk control was built to deliver, and the machinery is by now standard (Vovk et al., 2005; Bates et al., 2021; Angelopoulos et al., 2021). The standard recipe has exactly one lever: a threshold on a correctness score, calibrated on held-out data, above which we emit the model's answer and below which we abstain.

Applied to item 518, the recipe has three options, and all three are unsatisfying.

It can **emit** the answer. But $s = 0.18$ is low, and if we set the threshold low enough to admit this item we also admit a large population of items that genuinely are uncertain and genuinely are wrong; the risk budget is consumed and the guarantee fails.

It can **abstain**. This is what every conformal method for vision–language models we are aware of does, and it is safe. It is also, here, absurd: we are declining to answer a question to which a component already inside our own system supplies the correct answer with high confidence. We pay a coverage cost for an item we could have got right.

It can **delete the claim** — the claim-filtering move used in the generative setting (Li et al., 2025; Mohri & Hashimoto, 2024). In a yes/no setting deletion is abstention. In a captioning setting it is slightly better than abstention and considerably worse than what a reader wants, since a caption with a hole in it is not a caption that has been corrected.

What none of the three can do is emit **yes**. That option is not missing because it was tried and found wanting. It is missing because the framework does not contain it. The action space of a conformal selective predictor is $\{\text{emit the model's answer}, \text{abstain}\}$, and this paper is about what happens when a third action — *emit a different answer, and certify that too* — is added to it.

1.2 Two literatures that do not meet

The reason the fourth action is absent is, we think, sociological as much as technical: the two research communities with a stake in it work on different halves of the problem and rarely cite each other.

One literature makes the model wrong less often. This is by far the larger body of work, and it is inventive. It modifies decoding to penalise the attention patterns associated with over-trusting the language prior (Huang et al., 2024); contrasts logits from clean and corrupted views of the image (Leng et al., 2024; Zhang et al., 2025); localises the failure to particular layers and heads and intervenes surgically (Jiang et al., 2025; Wan et al., 2025; Zhao et al., 2025b; Deng & Yang, 2025; Cheng et al., 2026); injects the output of a vision expert during generation (Zhao et al., 2025a); verifies and rewrites after the fact (Yin et al., 2024; Zhou et al., 2024; Wu et al., 2025); or grounds the model in retrieved external knowledge (Lee et al., 2024; Xia et al., 2024). The measured effects are substantial and the methods are cheap. But the output of this literature is always of the form *our hallucination rate is lower than theirs on this benchmark.* A deployer who asks "what fraction of the claims you emit tomorrow will be false?" gets an empirical average over a fixed test set, which is not an answer to the question asked.

The other literature answers that question, and charges for it. Conformal prediction and its risk-control descendants turn any heuristic score into a decision rule with a finite-sample, distribution-free guarantee. Applied to generation, the result is *selective prediction*: emit when a correctness score clears a calibrated threshold, abstain otherwise, and the error rate *among emitted outputs* is at most α with probability $1 - \delta$ (Quach et al., 2024; Mohri & Hashimoto, 2024; Li et al., 2025; Lee et al., 2025). This is exactly the promise a deployer wants, it holds without distributional assumptions beyond exchangeability, and it holds for the next batch and not merely the last one. The price is paid in coverage: every item the rule cannot vouch for is an item nobody is served.

Neither literature is wrong. But notice what falls between them. The first literature builds systems that can *repair* a wrong answer — Woodpecker rewrites captions with detector output, REVERSE resamples flagged spans, MARINE injects object-level guidance — and never certifies the result. The second literature certifies rigorously and never repairs. The obvious composition, *certified repair*, is what the verify-and-correct line (Yin et al., 2024; Zhou et al., 2024; Dang et al., 2026) gestures at without achieving: pipelines whose correctness rests on benchmark averages. This paper asks the question that line leaves open — what would it take to attach a real guarantee to a system that changes its own answers?

1.3 The price of a guarantee, and why it is not a tuning problem

Before asking that, it is worth being precise about how expensive abstention is, because the size of the number is what makes the fourth action interesting rather than merely tidy.

Kotte (2026a) give the answer in closed form. If the base risk of the underlying model is μ and the target is $\alpha < \mu$, then *any* conformal selective predictor must abstain on at least

$$\frac{\mu - \alpha}{1 - \alpha}$$

of inputs. The bound does not depend on the quality of the score, the tightness of the concentration inequality, or the size of the calibration set. A perfect oracle score obeys it. A larger calibration set does not relax it. It is a property of the action space, not of the estimator.

The consequences on real systems are not subtle. On our POPE-1500 setup with LLaVA-1.5-7B the empirical base risk is $\mu \approx 0.18$; at $\alpha = 0.10$ at least 8.9% of inputs must be refused before any method is chosen. Our own tuned filtering pipeline, which is competitive with published conformal VLM results, certifies 42.1% coverage on POPE-adversarial at $\alpha = 0.10$: three in five questions go unanswered. On HallusionBench, where the base risk is high enough that the floor swallows the entire budget, it certifies 0.0%. Zero. The guarantee holds vacuously, over an empty emitted set, which is exactly the failure mode a practitioner would call useless.

This is the point at which the natural engineering instinct — *find a better score* — runs into the theorem. A better score moves you toward the floor faster. It cannot move you through it. Whatever is to be done must be done to the action space.

1.4 Re-reading the premise

So we re-read the premise. Kotte (2026a) state it plainly: the predictor may only emit the model's own answer or abstain. Every conformal method for VLMs we surveyed satisfies it, and satisfies it structurally rather than incidentally. Claim-filtering deletes unsupported claims (Li et al., 2025; Mohri & Hashimoto, 2024). Abstention methods decline (Lee et al., 2025; Wagner, 2026). The closest concurrent work, budgeted conformal evidence acquisition (Xu et al., 2026), acquires *additional visual evidence* for borderline claims and re-scores the original assertion — a genuinely clever move that buys real coverage, and one that explicitly leaves the emitted answer unchanged. In every case the object emitted is the model's output or nothing.

Item 518 says that this is a choice, not a necessity. Our system already contains a component that knows the right answer. The question is whether we can emit that answer *and still have a guarantee.*

What changes. Permitting substitution does not evade the mathematics; it changes the object being certified. If the emitted answer may originate outside the model, the quantity to control is the risk of a *mixture* — accepted model answers together with substituted ones — and the bound of Kotte (2026a) no longer binds, because its premise is violated. The guarantee must be re-derived for the mixture, which we do (Theorem 1), and the substitutions must come from somewhere trustworthy, which turns out to be a hard constraint rather than an implementation detail (§4.4).

Why it can win, in one sentence. High-precision repairs are *low-risk* emissions. Adding them to the emitted set pulls its average risk below α, which slackens the binding constraint, which lets the acceptance threshold move down, which admits a much larger population of ordinary accepted items. The gain is therefore levered: in our experiments 0.9–1.8% of items repaired buys 1.9–4.7 percentage points of certified coverage, a 1.9–3.4× multiplier. The mechanism is not "we fixed some answers"; it is "fixing a few answers changed the price of all the others".

1.5 What went wrong on the way, and what it taught us

It would be convenient to report that the idea worked. It works in four settings out of five, and the interesting content of this paper is largely in the failures, each of which turned out to be forced rather than accidental. We list them here because they are the spine of the argument, and each is a section below.

The cheap version defeats itself. The first design certified repairs using the complement of the score that flagged the original: if s is low, surely $1 - s$ vouches for flipping the answer. This is circular, and we can say exactly how. For binary decisions the risk of the flipped answer on a flagged region equals the model's *accuracy* on that region, so a self-repair region is cheap only where the model is right less than α of the time — a strictly stronger demand than finding a region where it is unconfident. We first thought this impossible and can now be more precise: it is admissible whenever the accepted region has slack (Proposition 2), but on the settings where correction is worth doing the region does not exist (20.9% accuracy in the bottom 2% of our score against $\alpha = 0.10$; 64.3% on Qwen2-VL), and attempting it makes the fixed sequence abort outright on 65–100% of splits. So a second, independent channel is required in practice, which is the paper's main structural claim — an empirical regularity with a mechanism, not a theorem.

The obvious parameterisation loses on half our backbones. Given two channels, the natural move is to sweep both gates with the single risk-control parameter. We did, and it lost 4.2 points on Qwen2-VL and 2.9 on VCD-decoded LLaVA relative to plain filtering, because a liberal acceptance threshold drags the repair gate open with it and admits low-precision repairs precisely when the budget is tightest. The fix is to decouple: the repair gate is fixed and strict, and only acceptance is calibrated (§4.5). This is a small change with a large effect, and we report the version that failed because the reason it failed is the reason the final design looks the way it does.

The gains are real, modest, and not uniform. Averaged over settings the improvement is a few points of certified coverage, with one reversal. We chased the reversal and found a partial explanation: the achievable gain is capped by the abstained mass, so a benchmark that already certifies most of its inputs has little to gain and the same exposure to loss. This yields a checkable diagnostic — μ comfortably above α, with an applicable evidence channel — which is consistent with the failure on AMBER-discriminative but, being an upper bound on the gain, does not predict it (§4.7, Remark 4). Upside is bounded; downside is not.

Most of our headline margin is not our algorithm. Compared against a published conformal VLM scorer we gain about 67 points of certified coverage on one setting. Decomposed, roughly 62.6 of those points come from the correctness score — a boring combination of logit and grounding features — and around 4.4 from the certified-correction mechanism itself. We report the decomposition rather than the headline (§5.5), and we would rather a reader trust the 4.4 than be sold the 67.

In generation, correction backfires. The motivating case for repair is generative: replacing a hallucinated object word is plainly more useful than deleting the sentence containing it. Extending the guarantee to a sequence requires that repairing a claim not make later claims worse — a monotonicity property that would license a fixed-point argument. We tested it directly with a paired matched-prefix design and it fails: repaired prefixes yield $+0.207 \pm 0.091$ additional hallucinated mentions downstream ($n = 121$, $p = 0.025$). The fixed-point route — which needs the correction map to be monotone — is closed. We give the weaker coupling bound that survives (Theorem 2) and report the failure as a finding, since it constrains anyone attempting on-policy certified editing.

The closest competitor turns out to be an ally. Concurrent work (Xu et al., 2026) shares our benchmarks, our machinery, and our motivation, and we initially read it as a scoop. Reproducing it under a common base score shows the two mechanisms rescue disjoint populations of claims — theirs the ones where more looking resolves the doubt, ours the ones where the model is simply wrong and something else is right — and compose additively (§5.5). We narrow our novelty claim accordingly, to the substitution action and its guarantee.

We report all of this. The paper is arranged so that each design decision is forced by a measurement rather than asserted, and the negative results appear as findings rather than as caveats appended to a positive headline.

1.6 Contributions

1. **A procedure and its guarantee** (§4). CCRC emits accepted, repaired, or abstained outputs and certifies the risk of the emitted mixture: $\Pr[\mathcal{R}(\mathcal{E}) \leq \alpha] \geq 1 - \delta$ (Theorem 1), using fixed-sequence testing over a nested policy family with exact binomial bounds. We also give the minimum certifiable sample size (Lemma 1), a small result with practical teeth: a fixed sequence started below it aborts and returns zero coverage.

2. **Why repair cannot be self-certified in practice** (§4.4). Certifying a substitution using the complement of the score that flagged the original makes the repair's risk equal to the model's *accuracy* on the flagged region. Proposition 2 bounds the mass such a repair may occupy by the accepted region's slack; we then measure that on the settings where correction is worth doing no sufficiently low-accuracy region exists, and that the fixed-sequence test aborts entirely on 65–100% of splits when self-repair is attempted. The conclusion — use a second, independent channel — is an empirical regularity with a mechanism, not a theorem.

3. **A mechanism and a precondition** (§4.6, §4.7). We show why repair helps when it helps: high-precision repairs lower the emitted set's average risk, relaxing the binding constraint and permitting a more permissive acceptance threshold (Proposition 3). The effect is levered: 0.9–1.8% repaired mass yields 1.9–4.7 points of coverage. We then bound the achievable gain by the abstained mass (Proposition 4), which explains the asymmetry we observe — upside is capped, downside is not — and yields a diagnostic that is consistent with, though it does not predict, the one benchmark where CCRC loses.

4. **A sequential negative result** (§6). In a paired matched-prefix experiment on free-form captions, repairing a hallucinated claim significantly increases hallucination in the continuation. This rules out the *correction-map* monotonicity on which a fixed-point argument for an on-policy sequential guarantee would rest (a different property from risk monotonicity in λ, which Theorem 1 never uses), and we give the weaker coupling bound that survives (Theorem 2).

5. **Honest attribution and positioning** (§5.5). We decompose our margin over a published scorer and report that most of it is the score, not the procedure. We reproduce the closest concurrent method (Xu et al., 2026) and find the two mechanisms compose rather than compete.

1.7 Roadmap

§2 situates the work between the two literatures above. §3 fixes notation and derives the abstention floor in our own terms, so that the object CCRC certifies can be compared with it directly. §4 gives the procedure, its guarantee, the impossibility result, the decoupling argument, the dilution mechanism, and the precondition. §5 reports experiments across five benchmarks, two backbones and one mitigation decoder, including the setting where the method fails and the baselines that narrow the credit due to it. §6 reports the sequential falsification and what survives it. §7 isolates the score combiner (with a validity trap we fell into), the choice of concentration bound, and positioning against prior art. §8 states the limitations we consider load-bearing.

2 Related work

This section is organised around the gap identified in §1. We first survey what is known about hallucination in vision–language models and the very large body of work that reduces it (§2.1), noting throughout that essentially none of it produces a guarantee. We then survey distribution-free risk control (§2.2) and its application to language and vision–language generation (§2.3), noting that essentially all of it operates by deletion or abstention. §2.4 covers non-conformal uncertainty quantification, which supplies many of the scores both literatures use. §2.5 treats the two concurrent papers that bear most directly on our claim, and §2.6 the shift induced by acting on one's own output. §2.7 states what, after all of this, we take to be new.

2.1 Hallucination in vision–language models

Phenomenon, taxonomy and causes. Hallucination in multimodal models is conventionally decomposed by what is fabricated: *existence* (an absent object is asserted), *attribute* (a present object is described wrongly), and *relation* (spatial or interactional claims that do not hold) (Wang et al., 2023a). Surveys of the multimodal case (Bai et al., 2024; Liu et al., 2024b) and of the text-only antecedent (Ji et al., 2023) converge on a small set of proximate causes. First, *language-prior dominance*: the autoregressive decoder is trained on text in which certain objects co-occur, and at generation time it will complete "a desk with a keyboard and a. . . " with `mouse` whether or not a mouse is visible. Second, *visual under-attention*: attention mass on image tokens decays with position, so later tokens in a long caption are generated with progressively less visual grounding. Third, *instruction-tuning artefacts*: supervision built from captions that describe more than is visible teaches the model to over-describe (Liu et al., 2024a). Fourth, ordinary *miscalibration* of the underlying classifier (Guo et al., 2017), which matters for us because it determines how informative the model's own logits are as a correctness score.

The taxonomy is not decoration; it determines which evidence can refute a claim, and therefore which claims a system like ours can repair. Existence claims are checkable by an open-vocabulary detector (Minderer et al., 2023; Liu et al., 2024d). Attribute claims require finer perception than a detector supplies. Relation claims require structure that neither a detector nor a similarity model represents. We return to this in §5.3, where the composition of a benchmark — not its difficulty — decides whether our evidence channel applies at all.

Benchmarks, and what they measure. POPE (Li et al., 2023) probes existence with balanced yes/no questions under three negative samplings: *random* objects, *popular* (frequent in the corpus), and *adversarial* (frequently co-occurring with objects that are present). The last is substantially harder precisely because it pits the language prior against the image, which makes it the most informative split for our purposes. CHAIR (Rohrbach et al., 2018) scores free-form captions by the fraction of mentioned objects absent from MS-COCO ground truth, and remains the standard generative metric despite its restriction to a closed vocabulary. AMBER (Wang et al., 2023a) provides an LLM-free benchmark with per-image lists of present and hallucinated objects across generative and discriminative splits, and annotates attribute and relation types as well as existence. MMHalBench (Sun et al., 2024a) evaluates diverse hallucination types in question answering using a strong judge model. MME (Fu et al., 2023) measures broad perception and cognition across fourteen subtasks, only some of which are grounding-checkable. HallusionBench (Guan et al., 2024) targets visual illusions, charts, and reasoning traps, and is hard enough that strong models approach chance — a property that turns out to matter a great deal for any method with an abstention floor. GQA (Hudson & Manning, 2019) supplies compositional questions over scene graphs.

We use POPE, AMBER, MME, GQA and HallusionBench. One methodological remark belongs here rather than in the experiments: single-type AMBER subsets are label-degenerate (the `hallucination` subset has all-`no` gold answers, the `relation` subset all-`yes`), which inflates AUROC to 1.0 for any score correlated with the question type. We report on the mixed subset for this reason, and flag it because the degeneracy is easy to miss.

Mitigation by decoding. The largest and most active family modifies the decoding distribution at inference time, with no retraining. OPERA (Huang et al., 2024) identifies an "attention sink" pattern characteristic of over-trust in the language prior, penalises it in the beam score, and retrospectively reallocates attention. Visual contrastive decoding (Leng et al., 2024) contrasts logits computed from the clean image against logits from a diffusion-corrupted version, subtracting the component of the distribution that survives visual degradation — i.e. the component driven by the prior. Instruction-contrastive and focal-contrast variants use other perturbations of the input (Chen et al., 2024b), and Favero et al. (2024) formalise the contrast as an information-grounding objective. DeGF (Zhang et al., 2025) closes a generative loop: it synthesises an image from the candidate caption and diffs it against the input. MARINE (Zhao et al., 2025a) injects object-level guidance from an external vision expert into the decoding step, which is the closest decoding-time analogue of our evidence channel — with the difference that it steers the model rather than replacing its answer, and offers no bound.

These methods are cheap, training-free, and demonstrably effective. Their relation to our work is compositional rather than competitive: they lower the base risk μ, while we control the risk of what is emitted. Because the abstention floor is $(\mu-\alpha)/(1-\alpha)$, *any* reduction in μ directly relaxes the constraint we operate under, so the two lines of work multiply rather than substitute. We verify this empirically by running our layer on top of Leng et al. (2024) in §5.4, and find both that the composition works and that the coupled variant of our own method fails there — which is how we discovered the need to decouple the repair gate.

Mitigation by attention-level and mechanistic intervention. A related line localises hallucination to specific components and intervenes there. Attention-lens style analyses trace hallucinated tokens to identifiable heads and layers (Jiang et al., 2025; Wan et al., 2025), and interventions range from masking (Deng & Yang, 2025) through layer-selective contrastive readout (Zhao et al., 2025b) to causal gating of the information paths implicated (Cheng et al., 2026). These give the most detailed available picture of *where* the failure lives. They are orthogonal to our concern, which is not where the error comes from but what the emitted set's risk is once it exists; we cite them because they are the strongest current evidence that hallucination is a structured, localisable phenomenon rather than noise — which is in turn why an independent channel can be expected to be informative about it.

Mitigation by training and preference optimisation. A second family changes the model. Factually augmented RLHF (Sun et al., 2024b) scores responses against ground-truth image content during reward modelling. RLHF-V (Yu et al., 2024) collects fine-grained *correctional* human feedback — humans edit hallucinated segments — and aligns on the edits, which is conceptually the closest training-time analogue of certified correction, minus the certificate. Robust instruction tuning (Liu et al., 2024a) attacks the artefact at its source by constructing negative instructions. These require training and produce a better μ; again, complementary.

Mitigation by post-hoc verification and rewriting. The third family is the one our work descends from: generate, then check, then change. Woodpecker (Yin et al., 2024) extracts object claims from a caption, queries detectors, and rewrites the caption using a language model. LURE (Zhou et al., 2024) identifies hallucination-prone captions from co-occurrence and uncertainty statistics and revises them. REVERSE (Wu et al., 2025) trains the model to emit explicit confidence markers during generation and then retrospectively resamples the spans it flagged, which makes it the closest work in *spirit* to ours: it also changes the output rather than deleting it. Knowledge-grounded variants replace detectors with retrieval over multimodal knowledge graphs (Lee et al., 2024) or domain corpora (Xia et al., 2024).

Every method in this family performs the same three steps we do — flag, consult external evidence, substitute — and none of them certifies the substitution. The quantity they report is a benchmark average after rewriting. Dang et al. (2026) is of exactly this type, and the present paper can be read as asking what it would take to attach a guarantee to such a pipeline. The attachment is not a matter of wrapping the pipeline in a calibration loop: the natural way to certify a correction is circular (§4.4), and the honest version requires an architectural commitment that the verify-and-rewrite literature happens to satisfy by accident rather than by design.

2.2 Conformal prediction and distribution-free risk control

From coverage to risk. Conformal prediction (Vovk et al., 2005; Papadopoulos et al., 2002; Lei et al., 2018) converts any score into prediction sets with finite-sample marginal coverage under exchangeability alone; see Angelopoulos & Bates (2023) for a modern treatment. Refinements adapt the sets to local difficulty (Romano et al., 2019), exploit leave-one-out structure (Barber et al., 2021), control error per class (Sadinle et al., 2019; Ding et al., 2023), condition on groups (Vovk et al., 2003), and relax exchangeability itself (Barber et al., 2023).

For our purposes the relevant generalisation is from *coverage* to *risk*. Risk-controlling prediction sets (Bates et al., 2021) bound the expectation of a monotone loss rather than a miscoverage indicator. Conformal risk control (Angelopoulos et al., 2024) extends this to a general family of monotone risks with a single scalar parameter. Learn-Then-Test (Angelopoulos et al., 2021) drops monotonicity altogether and treats the choice of configuration as a multiple-testing problem, which is the tool we need: our policy family is indexed by a threshold whose induced risk is *not* guaranteed monotone once repairs enter the emitted set, and LTT's fixed-sequence testing lets us search a nested family at full δ without a Bonferroni penalty.

The bound matters. Which concentration inequality supplies the per-hypothesis p-value has a large practical effect at the sample sizes available in VLM calibration. Hoeffding's inequality is loosest; empirical Bernstein adapts to observed variance; betting-style e-value constructions are tighter again (Waudby-Smith & Ramdas, 2024). For a binary loss — our case, since an answer is right or wrong — the exact Clopper–Pearson interval (Clopper & Pearson, 1934) dominates all of these, and we confirm the ordering experimentally in §7. Anytime-valid inference via e-processes and Ville's inequality (Waudby-Smith & Ramdas, 2024; Ramdas et al., 2023) extends validity to data-dependent stopping, which is the natural machinery for sequence-level guarantees and which we take up — and find insufficient on its own — in §6.

Selective prediction is older than conformal prediction. The accept/abstain trade-off dates to Chow (1970); its statistical theory (El-Yaniv & Wiener, 2010; Bartlett & Wegkamp, 2008; Cortes et al., 2016) and its deep-learning instantiations (Geifman & El-Yaniv, 2019) predate the conformal treatment. What conformal methods add is not the idea of abstaining but the distribution-free, finite-sample character of the promise. It is worth keeping the distinction in view, because the impossibility bound we build on is a statement about the selective-prediction *action space*, and therefore applies to the classical formulations just as much as to the conformal ones. Learning-with-rejection theory has, to our knowledge, likewise never considered a *reject-and-substitute* action with a risk certificate.

2.3 Conformal methods for language and vision–language models

Text. Conformal language modeling (Quach et al., 2024) calibrates sampling, stopping and rejection rules so that the returned set of generations contains an acceptable output with high probability. Conformal factuality (Mohri & Hashimoto, 2024) decomposes an output into sub-claims and *backs off* — progressively removing the least supported claims — until a factuality guarantee holds, which makes the emitted object a subset of the original. Conformal alignment (Gui et al., 2024) certifies which foundation-model outputs are trustworthy enough to use. Multiple-choice settings admit a direct application of split conformal prediction (Kumar et al., 2023), and conformal machinery has been used to calibrate when an LLM planner should ask a human for help (Ren et al., 2023) — an action space of {act, defer}, again without substitution. Token-level and next-token variants calibrate on entropy or logit statistics (Xu & Lu, 2025; Chen et al., 2025). Online and bootstrapped variants address feedback loops and estimator variance (Lee et al., 2025; Pang et al., 2025), and pipeline-level joint coverage over multiple stages has been studied (Kotte, 2026b).

Vision–language. ConfLVLM (Li et al., 2025) is the closest published system in the multimodal setting. It treats each generated detail as a hypothesis, scores it with a domain-appropriate heuristic — CLIP similarity (Radford et al., 2021) for natural-scene descriptions, BiomedCLIP for radiology, LayoutLMv3 for documents — and filters unsupported details, reporting a reduction in claim error from 87.8% to 10.0% for LLaVA-1.5 scene descriptions at a stated coverage cost. Wagner (2026) decompose the reliability question along a coverage axis and a risk axis and study the trade-off directly.

The common operation. Across this entire body of work — text and multimodal, offline and online, claim-level and sequence-level — the operation performed on a flagged claim is the same: **deletion, abstention, or back-off**. This is the empirical basis for our reading of Kotte (2026a): the premise of the impossibility bound is not an artefact of that paper's formalism, it is a faithful description of what every method in the area actually does. Nothing in the literature emits a claim the model did not produce and then certifies the result.

2.4 Uncertainty quantification without a guarantee

Our channel-1 score is not novel and we do not claim it is; it belongs to a large literature on estimating generative uncertainty, which supplies the raw material that conformal wrappers calibrate. Sequence likelihood and length-normalised logprobs are the baseline. Kadavath et al. (2022) show that models can be prompted to predict their own correctness. Semantic uncertainty (Kuhn et al., 2023) clusters samples by entailment before computing entropy, so that paraphrases do not count as disagreement, and semantic entropy scales this into a practical hallucination detector (Farquhar et al., 2024). Verbalised confidence (Lin et al., 2022) elicits a probability in words. SelfCheckGPT (Manakul et al., 2023) detects hallucination by sampling several generations and measuring mutual consistency, which requires no logit access; self-consistency (Wang et al., 2023b) uses the same signal to improve accuracy rather than to flag. Calibration of the underlying classifier (Guo et al., 2017) determines how much any of these are worth.

Two observations relevant to our design. First, all of these are *self-referential*: they interrogate the model about its own output. This is exactly the property that §4.4 shows to be fatal for certifying a correction — a score derived from the model cannot vouch for an answer that contradicts the model. Second, the detectors used in the mitigation literature (open-vocabulary detection (Minderer et al., 2023; Liu et al., 2024d), image–text similarity (Radford et al., 2021)) are not uncertainty estimates at all but independent perceptual evidence, and it is precisely their independence, not their accuracy, that makes them usable as channel 2.

2.5 The two closest works, and what separates us from them

The impossibility bound. Kotte (2026a) asks when conformal risk control can certify structured LLM outputs at all, and the paper is best read as three results with one practical moral. First, the impossibility result: when the base risk exceeds the target, any distribution-free method must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of examples. Because μ and α are both known before calibration, this doubles as a *closed-form feasibility test* — one can determine whether CRC will work at all without running it. Second, a certification hierarchy across Hoeffding, empirical Bernstein, and a betting-based e-CRC bound, with the Hoeffding-to-Bernstein step supplying the largest single gain (+37% certified configurations) and e-CRC earning its keep only when calibration data is scarce (10% certification at 20% of the data against 0% for Hoeffding). Third, a validation of adaptive conformal inference under cross-dataset shift, cutting risk-target violations from 71% to 21%, with the residual failures concentrated exactly where the impossibility bound predicts they must be. The empirical scope is substantial — six open-weight models from 3B to 72B parameters, eight datasets, four tasks, six nonconformity scores — and its verdict is blunt: hard NER, QA and classification configurations are simply uncertifiable at $\alpha = 0.10$.

Two things about this paper matter for us. The first is what it does *not* study: the setting is text-only structured generation (named-entity recognition, JSON extraction, question answering, classification), so the multimodal case, where an independent perceptual channel is available in a way it is not for pure text, is outside its scope. The second is the remedy it recommends. Faced with infeasibility at $\alpha = 0.10$, the prescription is to *relax the target*: at $\alpha = 0.30$–0.40 certification becomes practical (47% of NER, 40% of QA, 60% of classification configurations). That is a sound recommendation and, for many deployments, the right one. It is also an admission about the shape of the problem, and it is the fork this paper takes differently. We do not weaken or contradict the bound — it is correct, and we re-derive it in our own notation in Proposition 1. We exit its premise, and then pay for the exit by re-deriving validity for the mixture (Theorem 1) and by discovering the constraints that substitution imposes (§4.4 and Theorem 2).

The same author's pipeline-level conformal work (Kotte, 2026b) composes stage-wise guarantees and is subject to the same premise at every stage.

Three responses to one bound. It is worth naming the alternatives explicitly, because the contribution of this paper is best understood as a choice among them. Given a base risk μ above the target α, the floor $(\mu - \alpha)/(1 - \alpha)$ can be answered in exactly three ways. **(i) Relax the target.** Accept a weaker promise until it becomes feasible; this is the recommendation of Kotte (2026a), and at $\alpha = 0.30$–0.40 it works. **(ii) Lower μ, or sharpen the score.** Every mitigation method in §2.1 does the former; Xu et al. (2026) does the latter, by spending a budget to look again before deciding. Both stay inside the premise and both shrink the floor without removing it. **(iii) Change the action space.** Emit something the model did not say. This is the only response for which the theorem stops applying, and it is the one we take — at the cost of having to re-derive the guarantee and of inheriting two new constraints that (i) and (ii) do not face.

Budgeted conformal evidence acquisition. Xu et al. (2026) is concurrent, close, and worth describing carefully because a reader who skims both papers may reasonably ask what is left. Their diagnosis is the same as ours, and they state it more sharply than we do: to hold the hallucination rate below 5% on a balanced object-existence benchmark, a state-of-the-art conformal filter must abstain on more than 80% of claims. Their remedy is to replace the binary answer/abstain decision with a *three-way* choice — answer, abstain, or acquire additional visual evidence by re-examining the image (zooming, cropping, or applying a claim-specific intervention) under a bounded budget — then form a post-acquisition score and re-threshold, restoring validity by calibrating on post-acquisition rather than pre-acquisition scores. Their setting (POPE/COCO, LLaVA and Qwen backbones), their machinery (Clopper–Pearson p-values with fixed-sequence testing), and their motivation coincide with ours, and in the regime we reproduce they report certified coverage rising from roughly 28% to 37% at $\alpha = 0.10$.

The distinction is precise, and it survives the fact that both papers describe a "three-way choice". Their third action is *acquire*: spend budget to look again, and then answer or abstain on the strength of a better-informed score. Ours is *repair*: emit a different answer. BCEA changes *how confident it is* about the model's claim; we change *what claim is emitted*. In their own framing there is no corrected answer — an item that survives acquisition is emitted as the model wrote it, and an item that does not is abstained. This means BCEA remains inside the premise of Kotte (2026a): it moves items across the threshold by improving the score, which the bound explicitly permits, which is why it reduces the floor's bite without escaping the floor. It is, in the taxonomy below, the "better score" response rather than the "relax α" response. CCRC is the third response — change the action space — and it is the only one of the three for which the bound stops applying. The practical consequence is that the two rescue different populations: BCEA rescues claims where a closer look resolves genuine ambiguity, and CCRC rescues claims where the model is confidently or unconfidently wrong and an independent channel is right. We reproduce BCEA under a common base score and one mechanism per arm in §5.5 and find the gains approximately additive. We accordingly narrow our novelty claim to the substitution action and the guarantee over the resulting mixture, rather than to the observation that conformal VLM filtering abstains too much — an observation we share with them.

A note on the word "certified". The term is used in at least three senses in nearby literatures, and we mean the third. In certified robustness (Cohen et al., 2019) it denotes a worst-case guarantee over a perturbation ball. In automated program repair (Le Goues et al., 2019) a "validated" patch is one that passes a test suite, an empirical check with no statistical content. Our sense is the conformal one: a finite-sample, distribution-free, high-probability bound on the risk of the emitted mixture under exchangeability. We do not claim any per-item guarantee, and §8 is explicit about what the guarantee does not say.

2.6 Distribution shift caused by the predictor

Editing a generation changes the distribution of what follows it. This is not exogenous covariate shift, for which weighted conformal prediction is the standard remedy (Tibshirani et al., 2019), but *feedback* covariate shift: the shift is a function of the deployed decision rule itself (Fannjiang et al., 2022). Validity can be restored in some such settings by careful reweighting (Prinster et al., 2024), and adaptive conformal inference tracks drift online with no distributional assumptions (Gibbs & Candès, 2021). Model cascades

Table 1: Notation. Symbols are used with these meanings throughout; where a letter is reused in a different role we mark it explicitly at the point of use.

symbol	meaning
α	risk target for the emitted set
δ	failure probability of the certificate
μ	base risk of the underlying model, $\Pr[Y=0]$
$a^\star$, $\hat{a}(X)$	gold answer; the model's answer
Y	correctness *indicator* of the model's answer, $\mathbf{1}\{\hat{a}=a^\star\}$
$Y^{(2)}$	correctness indicator of channel 2's answer
$s(X)$	channel-1 correctness (conformity) score; large means likely correct
$m(X)$	channel-2 evidence margin
λ	acceptance permissiveness; larger λ accepts more
q	*fixed* repair-gate quantile, never calibrated
$Q_s(\cdot), Q_m(\cdot)$	empirical quantiles of s, m on the calibration fold
K_λ, V_λ	emitted count; error count among emitted
$\mathcal{R}, \mathcal{C}$	risk of, and coverage of, the emitted set
$k_{\min}$	smallest emitted count that can be certified (Lemma 1)
$\mathcal{I}$	sequential interference term (Theorem 2)

and routing systems face a structurally similar problem — the decision of which model answers changes the input distribution seen downstream (Chen et al., 2024a; Gupta et al., 2024).

Our sequential experiment (§6) shows that in generative correction this shift is not benign, and that the obstacle is stronger than the loss of exchangeability. Even granting a mechanism to restore validity under the induced shift, the intervention degrades the object being certified: repairing a hallucinated claim measurably increases hallucination in the continuation. Exchangeability can be repaired by reweighting; a monotonicity violation cannot.

2.7 What is new here

Positioned against the above, our claim is narrow and, we hope, correspondingly credible. The hallucination-mitigation literature repairs without certifying. The conformal literature certifies without repairing. Kotte (2026a) show that certifying without repairing has a hard price, and Xu et al. (2026) show how to reduce — but not escape — that price from inside the premise. What is new in this paper is (i) a selective predictor whose action space includes substitution and whose guarantee covers the resulting mixture, (ii) a budget constraint showing what self-certified substitution would cost, together with measurements showing that on the settings where correction pays there is no region cheap enough to satisfy it, so that a second independent channel is a practical requirement rather than a design preference, (iii) an account of the mechanism by which substitution buys coverage, with a checkable precondition for when it will not, and (iv) a falsification of the monotonicity property that a sequence-level version of the idea would need. Points (ii) and (iv) are negative, and we regard them as the more durable half of the contribution.

3 Problem setup

3.1 Selective prediction with a risk guarantee

Table 1 fixes notation. Two conventions are worth stating because nearby literatures differ: α is always the risk *target* and never a significance level, and δ is always the failure probability of the certificate.

Let $(X, a^\star)$ be a query–answer pair, where X contains an image and a question. A VLM produces $\hat{a}(X)$. Define the model's correctness indicator $Y = \mathbf{1}\{\hat{a}(X) = a^\star\}$ and the *base risk* $\mu = \Pr[Y = 0]$. A selective predictor is a measurable rule that either emits an answer or abstains. Writing $\mathcal{E}$ for the (random) set of emitted items,

$$\mathcal{R}(\mathcal{E}) = \Pr[\text{emitted answer wrong} \mid \mathcal{E}], \qquad \mathcal{C}(\mathcal{E}) = \Pr[\text{item emitted}].$$

The goal is a policy with $\mathcal{R}(\mathcal{E}) \leq \alpha$ certified at confidence $1-\delta$, and $\mathcal{C}(\mathcal{E})$ as large as possible. We assume throughout:

Assumption 1 (Exchangeability)**.** Calibration and test items are exchangeable draws from the same distribution.

Assumption 2 (Pre-specified policy family)**.** The candidate policies form a family $\{\pi_\lambda\}_{\lambda\in\Lambda}$ whose index set and ordering are fixed before seeing calibration data.

3.2 Filtering, and the floor it cannot cross

A *filtering* policy uses a score $s : \mathcal{X} \to [0,1]$, intended to increase with correctness, and emits $\hat{a}(X)$ when $s(X) \geq \tau$.

Proposition 1 (Abstention floor; Kotte 2026a, Prop. 3)**.** *Let $\mu > \alpha$. Any selective predictor whose emitted answer is always $\hat{a}(X)$ satisfies*

$$1 - \mathcal{C}(\mathcal{E}) \;\geq\; \frac{\mu-\alpha}{1-\alpha}$$

whenever $\mathcal{R}(\mathcal{E}) \leq \alpha$. The bound is independent of the score s, of the concentration inequality used, and of the calibration sample size.

Proof in our notation. Let $c = \mathcal{C}(\mathcal{E}) = \Pr[X \in \mathcal{E}]$ and split the total error mass by emission:

$$\mu \;=\; \Pr[Y=0] \;=\; \underbrace{\Pr[Y=0 \mid X \in \mathcal{E}]}_{=\mathcal{R}(\mathcal{E})}\, c \;+\; \Pr[Y=0 \mid X \notin \mathcal{E}]\,(1-c).$$

The second conditional probability is at most 1, and the emitted answer is always $\hat{a}(X)$, so no rule can make an abstained item's model answer correct — abstention removes the item from the emitted set but does not change Y. Hence $\mu \leq \mathcal{R}(\mathcal{E})\,c + (1-c)$. Imposing $\mathcal{R}(\mathcal{E}) \leq \alpha$ gives $\mu \leq \alpha c + 1 - c = 1 - c(1-\alpha)$, i.e. $c \leq (1-\mu)/(1-\alpha)$ and therefore

$$1 - c \;\geq\; 1 - \frac{1-\mu}{1-\alpha} \;=\; \frac{\mu-\alpha}{1-\alpha}.$$

Nothing in the argument refers to s, to the concentration inequality, or to the calibration size, which is the sense in which the bound cannot be engineered away. The single place it uses the premise is the step "no rule can make an abstained item's model answer correct": it is exactly this step that fails when the predictor may emit an answer other than $\hat{a}(X)$, and §4 is the consequence. $\square$

Two features of Proposition 1 drive this paper. It is *computable in advance*, since μ and α are known before any calibration; and it is *conditional on a premise* — that the emitted answer is the model's own — which is an assumption about the action space, not about the data.

3.3 Two channels

Definition 1 (Channels)**.** *Channel 1* is a correctness score $s(X) = \widehat{\Pr}[Y=1 \mid \phi(X)]$ fitted on features $\phi(X)$ derived from the model's own output distribution together with grounding features. Note the orientation: s is a *conformity* score, large for items the model is likely to get right, and acceptance is an upper-tail event $s \geq Q_s(1-\lambda)$. This is the mirror image of the usual nonconformity convention; we use it because it reads naturally as "confidence that the answer is correct". *Channel 2* is a separate predictor $b(X)$ that answers the query independently of the VLM, together with an evidence margin $m(X) \in [0,1]$ measuring how decisively it does so. We write $Y^{(2)} = \mathbf{1}\{b(X) = a^\star\}$ for channel 2's correctness. The only property the analysis uses is that $Y^{(2)}$ is not a deterministic function of Y; we do not assume statistical independence, and §8 notes that our instantiation violates it (channel 1's features include the detector score).

Remark 1 (Our instantiation repairs in one direction only). This is a property of our channel 2, not of CCRC, but it bounds every positive result below and we state it before the experiments rather than after. We instantiate b by thresholding an open-vocabulary detector's confidence $o(X)$ at $\theta = 0.15$ and set $m(X) = |o(X) - \theta|$. A maximally confident detector *negative* therefore attains only $m \leq \theta = 0.15$, whereas positives reach 0.85. The calibrated repair gate $Q_m(1-q)$ at $q = 0.10$ sits at 0.36, 0.37, 0.37 and 0.21 on our four settings — above θ in every case. Consequently 100% **of admitted repairs are "detector says the object is present"** (45/45, 60/60, 60/60, 23/23), and CCRC as evaluated can repair only *missed* objects. It never repairs the canonical false-positive existence hallucination that POPE-adversarial is built to elicit; those items are abstained, not corrected. This also explains why the repaired mass is only 0.9–1.8%. A symmetric margin with polarity-specific gates is the obvious remedy and we have not evaluated it.

In our instantiation channel 1 combines the VLM's yes/no probability, its confidence margin, and grounding features; channel 2 is an open-vocabulary object detector (Minderer et al., 2023) queried with the object named in the question, with m the absolute margin of its detection score around a fixed operating point. The essential property is that $Y^{(2)}$ is *not* a deterministic function of Y — the two channels can be right and wrong independently. §4.4 shows this is what makes repair possible.

4 Certified-correction risk control

4.1 The procedure

Fix a strict evidence quantile q (we use $q = 0.10$: repairs are drawn only from the top decile of channel-2 evidence). For permissiveness $\lambda \in [0, 1]$, policy π_λ acts as

$$\pi_\lambda(x) = \begin{cases} \textsc{accept } \hat{a}(x), & s(x) \geq Q_s(1-\lambda), \\ \textsc{repair with } b(x), & s(x) < Q_s(1-\lambda) \text{ and } m(x) \geq Q_m(1-q), \\ \textsc{abstain}, & \text{otherwise}, \end{cases} \tag{1}$$

where Q_s, Q_m are empirical quantiles computed on the calibration fold. Figure 1 shows the induced partition of the (s, m) plane. The emitted set is $\mathcal{E}_\lambda = A_\lambda \cup R_\lambda$ with A_λ the accepted and R_λ the repaired items, and the emitted-set error counts what is *actually emitted*,

$$V_\lambda = \sum_{i \in A_\lambda} (1 - Y_i) + \sum_{i \in R_\lambda} \big(1 - Y_i^{(2)}\big), \qquad K_\lambda = |A_\lambda| + |R_\lambda|. \tag{2}$$

4.2 Worked examples on real data

Before turning to aggregates, Figure 2 shows what the three decisions look like on individual POPE images, with the actual channel-1 score, channel-2 detection confidence, and resulting action for each. The three columns illustrate the regimes the procedure is designed to separate.

In the ACCEPT column the model answers a straightforward existence question confidently and correctly, the detector agrees, and the item is emitted unchanged; these cases carry the bulk of the coverage. The REPAIR column contains the cases that motivate the method: LLaVA is asked whether a computer mouse is present, answers "no" with $p = 0.21$, and is *wrong* — the mouse is there. Channel 1 correctly assigns a low correctness score ($s = 0.18$), which under a filtering policy would produce an abstention and no answer at all. The detector, however, localises the object with confidence 0.90, comfortably inside the top evidence decile, so CCRC substitutes the detector's answer and the item is emitted *correctly.* This is the dilution effect of §4.6 in a single instance: a high-precision repair converts an item that filtering would discard into a low-risk emitted item.

The ABSTAIN column shows the necessary complement. Here the model is wrong *and* the detector is also wrong, with weak evidence in both cases ($m = 0.04$ and $m = 0.03$); the questions ask about a handbag and a truck that are not present, and neither channel supplies a certifiable answer. Abstention is the correct action, and no amount of substitution would help. The existence of a substantial population of such items is why CCRC's gains are bounded by Proposition 4 rather than being able to recover the entire abstained mass.

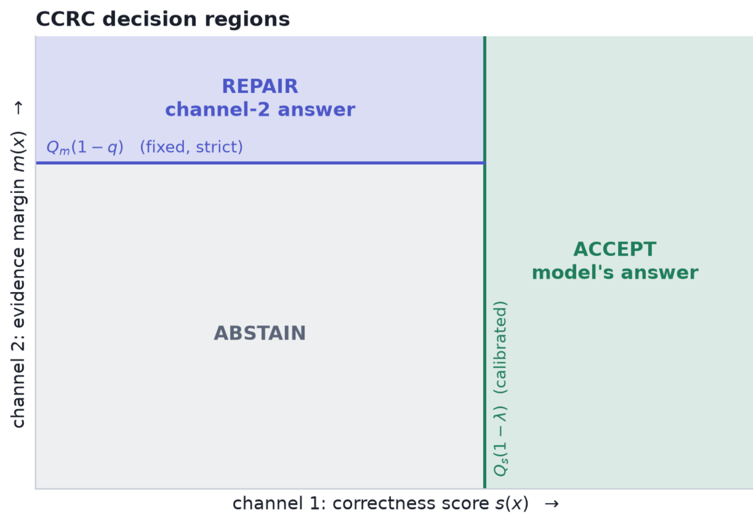

Figure 1: CCRC partitions the (s, m) plane. The acceptance threshold $Q_s(1-\lambda)$ is calibrated; the repair gate $Q_m(1-q)$ is fixed in advance and strict, so that as λ loosens the accepted region grows and the repair region shrinks. Keeping q fixed is what makes the family nested and lets a single fixed-sequence test suffice (§4.5).

4.3 Validity

Theorem 1 (CCRC validity)**.** *Let Assumptions 1–2 hold, let $\{\lambda_1 < \cdots < \lambda_M\}$ be a pre-specified grid, and let*

$$U(e, k; \delta) = \sup\{p : \Pr[\mathrm{Bin}(k, p) \le e] > \delta\}$$

be the one-sided Clopper–Pearson upper confidence bound. Define $\hat{\lambda}$ by fixed-sequence testing: proceed through $j = 1, 2, \ldots$ while $U(V_{\lambda_j}, K_{\lambda_j}; \delta) \le \alpha$ on the calibration fold, and set $\hat{\lambda}$ to the last index that passed. Then the deployed policy $\pi_{\hat{\lambda}}$ satisfies

$$\Pr\big[\mathcal{R}(\mathcal{E}_{\hat{\lambda}}) \le \alpha\big] \;\ge\; 1-\delta.$$

Proof. Fix j and consider the null $H_j : \mathcal{R}(\mathcal{E}_{\lambda_j}) > \alpha$. Conditional on $K_{\lambda_j} = k$, the emitted-set errors in (2) are, under Assumption 1, exchangeable Bernoulli indicators; note that this holds for the mixture because each emitted item contributes exactly one indicator, namely $1 - Y_i$ if accepted and $1 - Y_i^{(2)}$ if repaired, and membership in A_{λ_j} or R_{λ_j} is determined by (s, m) and the pre-specified λ_j, q rather than by the labels.

Exchangeability alone does not give $V_{\lambda_j} \sim \mathrm{Bin}(k, \mathcal{R})$, and the gap is not academic in our implementation: $Q_s(1-\lambda)$ and $Q_m(1-q)$ are empirical quantiles of the same fold whose errors are counted, so conditional on the calibration features the emitted indicators have item-specific means and V_{λ_j} is *Poisson binomial* with mean $k\bar{p}$, $\bar{p} = \mathcal{R}(\mathcal{E}_{\lambda_j})$, rather than binomial. Clopper–Pearson survives the substitution. The lower tail of a Poisson binomial with mean $k\bar{p}$ is dominated by that of $\mathrm{Bin}(k, \bar{p})$ (Hoeffding, 1956), i.e. $\Pr[V_{\lambda_j} \le v] \le \Pr[\mathrm{Bin}(k, \bar{p}) \le v]$ for $v \le k\bar{p}$; and since $U(v, k; \delta)$ is decreasing in v, the certification event $\{U(V_{\lambda_j}, k; \delta) \le \alpha\}$ is a lower-tail event on V_{λ_j}, so under H_j its probability is at most its probability under the equal-mean binomial. Hence, by construction of the Clopper–Pearson bound, $\Pr[U(V_{\lambda_j}, k; \delta) \le \alpha \mid H_j] \le \delta$: the event that the bound certifies a policy whose true risk exceeds α has probability at most δ, so $p_j = \mathbf{1}\{U \le \alpha\}$ is a valid level-δ test of H_j.

The index set and its ordering are pre-specified by Assumption 2, which is all that fixed-sequence testing requires for validity. Nesting is not needed for validity, but it is what makes the procedure powerful: because q is fixed, A_λ increases and R_λ decreases in λ, so $\mathcal{E}_\lambda = A_\lambda \cup M$ with $M = \{m \ge Q_m(1-q)\}$ fixed, and *coverage* is monotone increasing in λ even though the risk is not — which is what makes "the last index that passed" the right thing to return. Fixed-sequence testing applies: testing $H_1, H_2, \ldots$ in the pre-specified

Figure 2: Real POPE images with the actual quantities CCRC uses. Each panel reports the question, the ground truth, LLaVA's answer with its probability, the detector's answer with its confidence and evidence margin, the channel-1 correctness score, and the resulting decision. **Accept** (left): both channels agree and the model is right. **Repair** (centre): the model misses an object that is genuinely present and receives a low correctness score, while the detector localises it with high confidence, so the answer is substituted and the error is fixed — items a filtering policy would abstain on. **Abstain** (right): the model is wrong and the detector is also wrong with weak evidence, so no certifiable answer exists and the procedure correctly declines. Thresholds are the calibrated $Q_s(1-\lambda)$ and the fixed gate $Q_m(1-q)$ from one split.

order and stopping at the first non-rejection controls the family-wise error rate at δ without any multiplicity correction (Angelopoulos et al., 2021). The returned $\hat{\lambda}$ is among the rejected hypotheses, so with probability at least $1-\delta$ every rejected H_j is false, i.e. $\mathcal{R}(\mathcal{E}_{\hat{\lambda}}) \leq \alpha$. □

Remark 2 (why the mixture is not a complication)*.* The proof needs only that each emitted item contributes one $\{0,1\}$ error indicator whose distribution does not depend on the labels of other items. Whether that indicator refers to the model's answer or to channel 2's is irrelevant. This is precisely why Proposition 1 does not bind: its premise restricts the *action space*, which the guarantee itself does not require.

Lemma 1 (Minimum certifiable emitted-set size)**.** *With zero observed errors, $U(0,k;\delta) = 1-\delta^{1/k}$. Hence no emitted set of size k can be certified at level α unless*

$$k \;\geq\; k_{\min} = \frac{\ln \delta}{\ln(1-\alpha)}.$$

For $\alpha = \delta = 0.10$, $k_{\min} = \lceil 21.85 \rceil = 22$.

Proof. $\Pr[\mathrm{Bin}(k,p) \leq 0] = (1-p)^k$, so the upper bound with $e = 0$ solves $(1-p)^k = \delta$, giving $U(0,k;\delta) = 1-\delta^{1/k}$. This is decreasing in k, and $1-\delta^{1/k} \leq \alpha \iff \delta^{1/k} \geq 1-\alpha \iff k^{-1}\ln\delta \geq \ln(1-\alpha)$; dividing by the negative quantity $\ln(1-\alpha)$ reverses the inequality and yields $k \geq \ln\delta/\ln(1-\alpha)$. Since $U(e,k;\delta) \geq U(0,k;\delta)$ for $e \geq 0$, no error count can do better. □

Lemma 1 is elementary but consequential: a fixed sequence whose first policy emits fewer than $k_{\min}$ items fails at step one and, because fixed-sequence testing stops at the first non-rejection, returns *zero* coverage. We encountered exactly this failure and now begin the grid at the analytically derived minimum.

4.4 Why repairs cannot be self-certified in practice

The most economical design would avoid a second channel: if s says the model is probably wrong, emit the opposite. This section shows why that fails. The argument has to be made at the level of the *mixture*, because that is what CCRC certifies — a repair region whose own risk exceeds α is not automatically inadmissible, since a compliant blend can absorb it (Proposition 3). The correct statement is therefore not that self-repair is impossible but that it is *self-defeating*: the mass it can buy is bounded by the accepted region's slack, and the fixed-sequence test aborts before it can be collected.

Proposition 2 (Self-repair is bounded by slack, and is not free)**.** *Suppose the answer space is binary and the repair is the negation of the model's answer, so that $Y^{(2)} = 1 - Y$. Let $R \subseteq \mathcal{X}$ be any repair region with mass c_r and let the accepted region have mass c_a and risk r_a. Then the risk of the repaired region is*

$$r_r \;=\; \Pr[\textit{repaired answer wrong} \mid X \in R] \;=\; \Pr[Y = 1 \mid X \in R],$$

i.e. exactly the model's ***accuracy*** *on R. Consequently, if $r_r > \alpha$ the mixture is compliant only for*

$$c_r \;\leq\; c_a \, \frac{\alpha - r_a}{r_r - \alpha}, \tag{3}$$

and a region of accuracy $\leq \alpha$ — for which no such constraint binds — requires the score to isolate inputs on which the model is right less than α of the time. That is strictly stronger than isolating inputs of low confidence*, and no monotone transformation of s can supply the difference.*

Proof. On R the emitted answer is $1 - \hat{a}$, which is correct exactly when $\hat{a}$ is incorrect; its error indicator is therefore Y, and averaging over R gives $r_r = \Pr[Y = 1 \mid X \in R]$. For the mixture, $\bar{r} = (c_a r_a + c_r r_r)/(c_a + c_r) \leq \alpha$ rearranges to $c_r(r_r - \alpha) \leq c_a(\alpha - r_a)$, which is (3) when $r_r > \alpha$ and is vacuous when $r_r \leq \alpha$. The final claim follows because any repair rule of the form $\{s(X) \leq t\}$ has $r_r = \Pr[Y = 1 \mid s(X) \leq t]$, a functional of the joint law of (s, Y) invariant to monotone reparameterisation of s. ☐

Proposition 2 converts an intuition — "a hallucination score reports suspicion, not knowledge of the truth" — into a budget. Two measurements show how little that budget buys.

No low-accuracy region exists where it is needed. Averaged over 60 fits of the canonical score, the model's accuracy in the bottom 2% of s is 20.9% on POPE-1500 and 44.6% on POPE-adversarial, against $\alpha = 0.10$; in the bottom decile it is 42.8% and 47.3%. On Qwen2-VL the bottom 2% is *above* chance at 64.3%, so flipping there is actively harmful. Only the two settings with the lowest headroom (VCD-decoded LLaVA at 2.1%, AMBER at 0.0%) contain a flippable region at all, and those are the settings with the least to gain.

The fixed sequence aborts before the slack can be spent. Substituting the negation of the model's answer at $q = 0.02$ and certifying the mixture exactly as in §4.3, certified coverage falls by 43.9, 34.7, 77.5 and 3.9 points on the four POPE settings. The mechanism is not a graceful trade: the fixed sequence returns *nothing at all* on 65%, 88%, 100% and 25% of splits respectively. The reason is structural. The self-repair region has fixed mass, so at the *start* of the grid — where the accepted set is smallest and K barely clears $k_{\min}$ — it is the largest fraction of the emitted set it will ever be, and its 20–64% error rate fails the very first hypothesis. Fixed-sequence testing must stop at the first non-rejection (§4.3), so the procedure never reaches the later λ at which (3) would in fact be satisfiable. Self-repair is defeated by the interaction between a fixed-mass, high-error repair block and the prefix structure of the test, not by the region being uncertifiable in isolation.

Table 2: Channel-2 (detector) accuracy by evidence-margin decile on POPE. Repairs are drawn only from the top decile, where accuracy is far above $1 - \alpha$. Accuracy degrades broadly with the margin, though not monotonically (the 8th decile falls slightly below the 6th), which is why the repair gate must be strict rather than merely above the median.

Margin decile	10th (top)	9th	8th	6th	1st (bottom)
Detector accuracy	1.000	0.960	0.927	0.934	0.513

Table 3: Sensitivity to the repair gate q: change in certified coverage relative to filtering, over four settings $\times$ two risk levels. q is fixed a priori, not tuned.

Repair gate q	Outcome across the eight cells	Worst cell
0.10 (top decile)	positive in all eight (+0.5 to +8.0)	+0.5
0.25	mixed	−16.3
0.50	mixed	−15.5

What this licenses us to claim. Not that certified correction is impossible without a second channel — (3) admits self-repair whenever the accepted region has slack and the flip region is clean enough, and on AMBER, where the bottom 2% has 0% accuracy, self-repair does gain (+1.3 points at $\alpha = 0.10$). What the measurements license is the weaker and sufficient claim that *on the settings where filtering is expensive enough for correction to be worth doing, no self-certifiable flip region exists*, so a second channel is required in practice. We state this as an empirical regularity with a mechanism, not as a theorem, and §8 lists it among the claims most likely to fail on a new benchmark.

Channel 2 escapes the budget in (3) because $Y^{(2)}$ is a genuinely different random variable, not a deterministic function of Y. Our detector is 93–100% accurate in its top evidence deciles (Table 2), so r_r there is far below α and the constraint is vacuous rather than binding.

4.5 Why the repair gate must be decoupled

An earlier version of the procedure tied the repair gate to λ, i.e. used $m(x) \geq Q_m(1 - \lambda)$ in (1). This is natural but harmful: as λ loosens to admit more accepted items, the repair gate loosens in step and begins admitting low-precision repairs, so the fixed sequence terminates earlier. Empirically the coupled version gained on LLaVA (+4.7, +5.3 points) but *lost* on Qwen2-VL (−4.2) and on a VCD-decoded model (−2.9). It also required certifying two families (with and without repair) at $\delta/2$ and selecting between them, wasting half the confidence budget.

Fixing q removes both problems. The family becomes nested in the single parameter λ, so one fixed-sequence test at full δ suffices (Theorem 1), and repair precision is held constant while λ sweeps. Table 3 reports sensitivity: at $q = 0.10$ the procedure gains in all eight POPE-family cells, whereas looser gates can lose badly.

4.6 Why it gains: dilution and leverage

Our initial hypothesis was that repairs *spend* the accepted set's unused risk budget: filtering typically realises a risk below α (we measure 8.0% at $\alpha = 0.10$), and repairs with error above α could be admitted while the blend stayed compliant. The measured effect runs the other way, and is stronger.

Proposition 3 (Dilution lowers the emitted risk pointwise in λ)**.** *Fix λ and let the accepted set have mass c_a and risk r_a, and the repair set mass c_r and risk r_r. The emitted risk is the mixture $\bar{r} = (c_a r_a + c_r r_r)/(c_a + c_r)$. If $r_r < r_a$ then $\bar{r} < r_a$, strictly decreasing in c_r, so every λ that is population-feasible for filtering remains population-feasible with repairs and further λ may become feasible. If instead $r_r > \alpha > r_a$, admitting repairs consumes slack and may render feasible λ infeasible.*

Proof. $\bar{r} - r_a = c_r(r_r - r_a)/(c_a + c_r)$, which is negative iff $r_r < r_a$, and $\partial\bar{r}/\partial c_r = c_a(r_r - r_a)/(c_a + c_r)^2$ has the same sign. Filtering is the special case $c_r = 0$, giving the feasibility claim. The last statement is the same computation with the inequality reversed. □

Remark 3 (Why this does not immediately imply a more permissive $\hat{\lambda}$). Proposition 3 is a statement about the *population* risk at fixed λ, and two gaps separate it from the certified $\hat{\lambda}$ the procedure returns. First, $\hat{\lambda}$ is chosen by a Clopper–Pearson bound on realised counts, not by population feasibility: a single repaired item that happens to be wrong raises V_λ and can tighten $\hat{\lambda}$ even when $r_r < r_a$. Second, fixed-sequence testing stops at the first non-rejection, so a larger feasible *set* yields a later stopping index only if feasibility is a *prefix* of the grid — which is risk monotonicity in λ, exactly what §4.3 declines to assume. We therefore treat the coverage gain as an empirical claim (§5.2) for which Proposition 3 supplies the mechanism, not a corollary. §4.4 exhibits the failure mode this remark anticipates: there, feasibility is not a prefix and the test aborts at the first index.

The empirically relevant regime is the first. Because repairs are drawn from the top evidence decile, their risk is *lower* than that of the acceptance gate's *marginal* items — the items that determine whether the constraint binds. Repairs therefore act as ballast: a small high-precision mass pulls $\bar{r}$ down and lets λ open further. The result is leverage. Figure 3B plots it: 0.9–1.8% of items repaired buys 1.9–4.7 points of certified coverage — per setting, ratios of 3.4×, 2.4×, 2.8× and 1.9×.

4.7 When it helps: a checkable precondition

CCRC is not a free improvement. Its upside is bounded by what filtering leaves on the table, while its downside — perturbing a nearly-saturated constraint — is not symmetrically bounded.

Proposition 4 (Gain is capped by abstained mass). *Let $\mathcal{C}_{\text{filt}}$ be the coverage certified by filtering at level (α, δ) and $\mathcal{C}_{\text{CCRC}}$ that certified by CCRC. Then*

$$\mathcal{C}_{\text{CCRC}} - \mathcal{C}_{\text{filt}} \;\leq\; 1 - \mathcal{C}_{\text{filt}},$$

and $1 - \mathcal{C}_{\text{filt}} \geq (\mu - \alpha)/(1 - \alpha)$ by Proposition 1. No matching lower bound on $\mathcal{C}_{\text{CCRC}}$ holds, so the upside is bounded by the abstained mass while the downside is not.

Proof. Coverage is at most one, so the gain cannot exceed the abstained fraction $1 - \mathcal{C}_{\text{filt}}$; the floor inequality is Proposition 1 applied to the filtering policy. The asymmetry claim follows because Proposition 3 shows the emitted risk strictly increases when repairs are admitted with $r_r > r_a$, and Remark 3 shows this can move $\hat{\lambda}$ downward without bound. □

Remark 4 (Why this is a diagnostic and not a prediction). An upper bound on the gain cannot predict a loss, and we previously overstated this. Write $1 - \mathcal{C}_{\text{filt}} = (\mu - \alpha)/(1 - \alpha) + \varepsilon$. One might hope ε is small so that headroom $\mu - \alpha$ alone governs the achievable gain, but our own data refutes it: on AMBER the floor is 1.6% while $1 - \mathcal{C}_{\text{filt}} = 7.9\%$, so $\varepsilon = 6.3\%$, four times the floor. The usable statement is therefore the crude one — the gain cannot exceed the abstained mass, so a setting that already certifies most of its inputs has little to win and the same exposure to loss — and the relationship between headroom and realised gain is an empirical trend with visible exceptions (§4.7), not a law.

Proposition 4 predicts our one negative result. On AMBER's discriminative split, $\mu = 11.4\%$ against $\alpha = 10\%$: the floor is 1.6%, filtering already certifies 92.1%, and the theoretical headroom is at most 7.9 points. Measured, CCRC *loses* 7.5 points there. Figure 3A shows the gain tracking $\mu - \alpha$ across all settings. Both quantities are known before calibration, so the decision to deploy CCRC can be made a priori.

4.8 Algorithm

5 Experiments

Protocol. Every reported number uses a three-way disjoint split — fit the score, calibrate the threshold, test — with fixed-sequence testing and Clopper–Pearson bounds at $\delta = 0.10$, averaged over 20–100 random

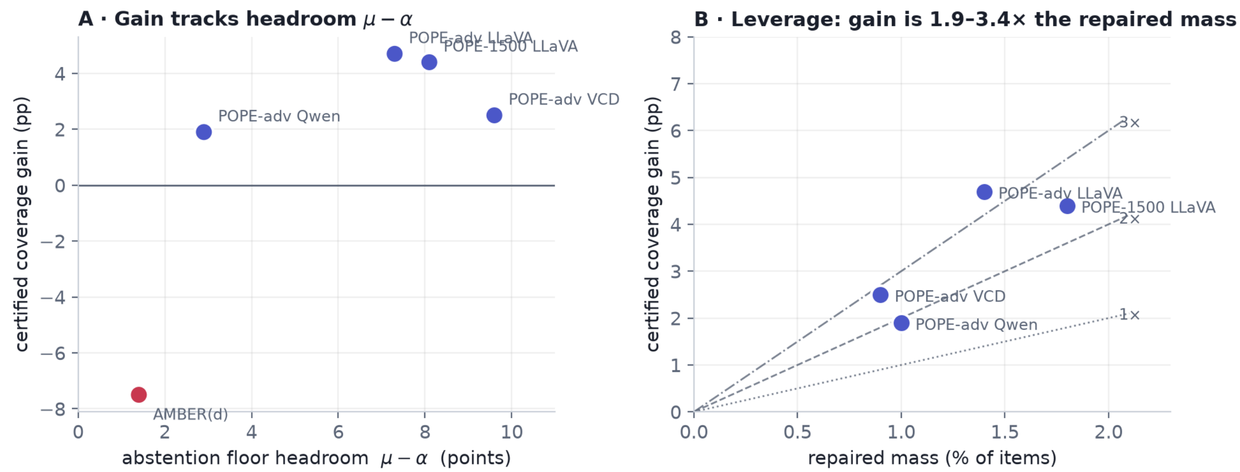

Figure 3: (A) Certified coverage gain against the abstention-floor headroom $\mu - \alpha$. Gains appear when the headroom exceeds roughly three points and reverse when it does not: AMBER(d), with 1.4 points of headroom, is the single loss. (B) Leverage: gain plotted against repaired mass, with 1×, 3× and 6× reference lines. A small high-precision repaired mass buys several times its own size in coverage, consistent with the mechanism of Proposition 3.

Algorithm 1 CCRC (calibration and deployment)

Require: folds $\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{cal}}$ (disjoint), levels α, δ, repair gate q, grid $\lambda_1 < \cdots < \lambda_M$ with λ_1 chosen so $K_{\lambda_1} \geq k_{\min}$ (Lemma 1; our implementation uses $\lambda_1 = \min\{0.9, \max\{0.05, 1.5k_{\min}/n_{\text{cal}}\}\}$, the 1.5 giving headroom against fold-to-fold variation in K)

fit channel-1 score s on $\mathcal{D}_{\text{fit}}$ ▷ disjoint from calibration; see §7

$t_m \leftarrow Q_m(1-q)$ on $\mathcal{D}_{\text{cal}}$ ▷ repair gate fixed

$\hat{\lambda} \leftarrow \varnothing$

for $j = 1$ to M **do**

  $A \leftarrow \{i \in \mathcal{D}_{\text{cal}} : s_i \geq Q_s(1-\lambda_j)\}$; $R \leftarrow \{i \notin A : m_i \geq t_m\}$

  $V \leftarrow \sum_{i \in A}(1 - Y_i) + \sum_{i \in R}(1 - Y_i^{(2)})$; $K \leftarrow |A| + |R|$

  if $K < k_{\min}$ **then continue** ▷ uncertifiable by Lemma 1; not evidence against H_j

  else if $U(V, K; \delta) \leq \alpha$ **then** $\hat{\lambda} \leftarrow \lambda_j$

  else break ▷ fixed-sequence testing stops at the first non-rejection

  end if

end for

at test time, for input x: **if** $\hat{\lambda} = \varnothing$ abstain on every input (coverage 0); **else if** $s(x) \geq Q_s(1 - \hat{\lambda})$ emit $\hat{a}(x)$; **else if** $m(x) \geq t_m$ emit $b(x)$; **else** abstain
▷ Q_s, Q_m are the calibration-fold quantiles fixed above, not recomputed at test time

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splits. The three-way split is not cosmetic: §7 shows that reusing the calibration fold to fit the score silently breaks the guarantee for high-capacity scores.

How validity is audited. Mean realised risk is *not* the guarantee. The claim is $\Pr[\mathcal{R} \leq \alpha] \geq 1 - \delta$, so the quantity to audit is the fraction of calibration draws whose risk exceeds α, compared against δ. We report both, and we distinguish two versions of the former, because realised risk on a finite test fold carries binomial noise that is not a validity failure. Scored on the test fold, exceedance is 11–15% on the POPE settings against $\delta = 0.10$; scored on the full item set — a lower-noise proxy for the population risk of the selected policy — it is 1–4%. The guarantee holds; the discrepancy is finite-sample noise in the audit, and we flag it because the naive statistic looks like a violation and the mean-risk statistic hides both.

Table 4: Selective prediction on POPE (LLaVA-1.5-7B, $n = 1500$). AURC is area under the risk–coverage curve (lower better). Realised test risk was $\leq \alpha$ in every row.

Score	AURC ↓	cov@10% ↑	AUROC ↑
Raw confidence	0.0775	62.1%	0.767
Learned, confidence only	0.0681	65.2%	0.786
+ CLIP grounding (Radford et al., 2021)	0.0632	67.5%	0.800
+ detection grounding (Minderer et al., 2023)	**0.0546**	**73.6%**	**0.835**
+ both	0.0538	74.9%	0.838

Table 5: Against a faithful self-consistency baseline on POPE-adversarial, the hardest split ($K = 3$ samples, LLaVA-1.5, $n = 450$).

Score	AURC ↓	cov@10% ↑
Self-consistency only	0.135	0.0%
Confidence + self-consistency	0.077	61.8%
+ detection grounding (ours)	**0.059**	**72.0%**

Models and data. Backbones are LLaVA-1.5-7B (Liu et al., 2024c) in 4-bit and Qwen2-VL-2B-Instruct (Wang et al., 2024); we additionally evaluate LLaVA decoded with visual contrastive decoding (Leng et al., 2024). Channel 2 is OWLv2-base (Minderer et al., 2023). Benchmarks are POPE (Li et al., 2023) (random and adversarial splits), AMBER discriminative (Wang et al., 2023a), MME (Fu et al., 2023), GQA (Hudson & Manning, 2019) and HallusionBench (Guan et al., 2024). All per-item extracted signals are released so that the analyses can be reproduced without a GPU.

5.1 The correctness score

Validity holds for any score; only efficiency depends on which one is used. We therefore first characterise what makes a score efficient here.

Table 4 compares scores on POPE. A learned combination of the model's own confidence signals is a solid baseline. Generic image–text similarity (Radford et al., 2021) adds little. An open-vocabulary *detection* signal adds a great deal: +8.4 points of certified coverage against +2.3 for CLIP.

Against a baseline given *both* confidence and multi-sample self-consistency — the scoring philosophy of Li et al. (2025) — detection grounding still adds 10.2 ± 8.6 points on the adversarial split (Table 5).

Two regularities follow. The advantage of grounding *grows as the guarantee tightens*: at $\alpha = 0.05$ it nearly doubles usable coverage (30.0% → 52.5%), while at $\alpha = 0.20$ both scores certify almost everything (Figure 4). And the advantage tracks how much the base model hallucinates: on the more accurate Qwen2-VL-2B the same addition is worth 1.9 points rather than 10.2. Grounding is most valuable exactly where selective prediction is most expensive.

Remark 5 (accuracy is not separability). A detector whose standalone accuracy matches the VLM's on POPE (83.1% vs. 81.9%) certifies far less coverage when used as a selective predictor in its own right (55.3% vs. 68.2%). Conformal coverage depends on how well a score *separates* correct from incorrect outputs, not on how often the underlying predictor is right. This also answers the natural objection to CCRC — "if channel 2 is good, why not just use it?" — in three of four settings (Table 8).

5.2 CCRC across settings

Table 6 reports CCRC against filtering under an identical protocol. The guarantee holds in every row. Gains are +0.5 to +8.0 points where the diagnostic of §4.7 is satisfied, and the single violation of that precondition is also the single loss.

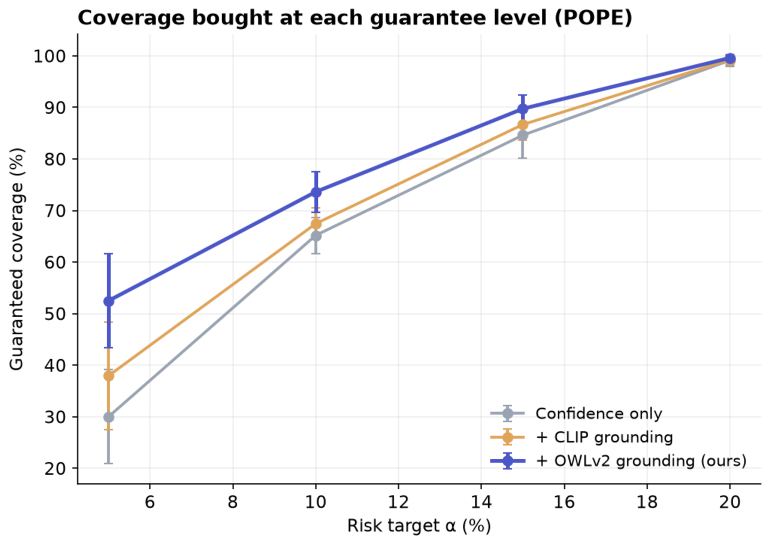

Figure 4: Certified coverage as a function of the risk target. The grounded score dominates at every level and the margin widens as α tightens; at $\alpha = 0.05$ it nearly doubles coverage. Error bars are one standard deviation over splits.

Table 6: CCRC (q=0.10) versus filtering. Realised test risk was $\leq \alpha$ in every row; realised risk for every row is reported in Table 14 and. The final row violates the precondition of Proposition 4 and is the only loss.

Dataset	Backbone	n	μ	α	Filter	CCRC	Gain
POPE	LLaVA-1.5	1500	18.1%	0.10	68.2%	**72.6%**	+4.4
POPE-adv	LLaVA-1.5	444	17.3%	0.10	42.1%	**46.8%**	+4.7
POPE-adv	Qwen2-VL-2B	591	12.9%	0.10	77.5%	**79.4%**	+1.9
POPE-adv	LLaVA+VCD	591	19.6%	0.10	45.0%	**47.5%**	+2.5
POPE	LLaVA-1.5	1500	18.1%	0.15	86.8%	**88.4%**	+1.7
POPE-adv	LLaVA-1.5	444	17.3%	0.15	69.2%	**77.2%**	+8.0
POPE-adv	Qwen2-VL-2B	591	12.9%	0.15	94.4%	**94.9%**	+0.5
POPE-adv	LLaVA+VCD	591	19.6%	0.15	73.4%	**75.4%**	+2.0
AMBER(d)	LLaVA-1.5	228	11.4%	0.10	92.1%	84.5%	−7.5

Isolating the cause of the AMBER loss. Two candidate explanations were tested and rejected. It is not a weak repair channel: the detector is 95.6% accurate on that dataset overall and 100% accurate in the top-decile region repairs are drawn from. It is not small sample size: subsampling POPE to the same $n = 228$ still gains +3.2 points, and the gain grows with n (+5.6 at 400, +10.3 at 700). The cause is the absence of headroom, exactly as Proposition 4 predicts.

5.3 Where grounding does not apply

Benchmark composition determines whether channel 2 exists at all. Table 7 reports the full sweep. On MME's full split, GQA's yes/no subset and HallusionBench, the questions are largely about attributes, counting or reasoning, so no object can be extracted for the detector to check; the procedure reduces to filtering, which remains valid. HallusionBench is instructive: LLaVA scores 51.2%, essentially chance, and the certified coverage is 0.0%. That is the guarantee working as intended — with $\mu \approx 0.49$ and $\alpha = 0.10$, Proposition 1 forces abstention on at least 43% of inputs, and in practice on all of them.

Table 7: Full dataset sweep. "grnd" indicates whether channel 2 applies. Coverage columns are confidence-only → with grounding, at $\alpha = 0.10$.

Dataset	Backbone	acc	grnd	cov@10%
POPE (1500)	LLaVA-1.5	81.9%	yes	63.6 → **73.1**%
POPE-adv	LLaVA-1.5	82.7%	yes	46.6 → **57.4**%
POPE-adv	Qwen2-VL-2B	87.3%	yes	83.5 → **85.5**%
POPE-adv	LLaVA+VCD	80.3%	yes	21.8 → **53.9**%
MME-existence[a]	LLaVA-1.5	95.0%	yes	underpowered
AMBER(d)	LLaVA-1.5	78.6%	partial	see Table 6
MME (full)	LLaVA-1.5	68.9%	no	5.7%
GQA (yes/no)	LLaVA-1.5	72.4%	no	17.8%
HallusionBench	LLaVA-1.5	51.2%	no	0.0%

[a] Only 60 items, leaving a 20-item calibration fold under the three-way split, which is below $k_{\min} = 22$ (Lemma 1). Sample size is not the only obstruction: at this subset's low base error rate $U(3, 60; \delta) = 0.108 > \alpha$, so power would bind even with the full 60 items available for calibration.

Table 8: Detector-only selective prediction versus CCRC ($\alpha = 0.10$, identical protocol).

Setting	VLM acc	det. acc	Filter (VLM)	CCRC	Detector only
POPE-1500 LLaVA	81.9%	83.1%	68.2%	**72.6%**	55.3%
POPE-adv LLaVA	82.7%	82.4%	42.1%	**46.8%**	13.0%
POPE-adv Qwen2-VL	87.1%	82.9%	77.5%	**79.4%**	21.8%
AMBER(d) LLaVA	88.6%	95.6%	92.1%	84.5%	**92.6%**

5.4 Composition with a mitigation decoder

Because the two programmes of §2 are orthogonal — mitigation lowers μ, risk control governs what is emitted — our layer should compose with a mitigation decoder. It does, and most strongly where the decoder's own confidence is poorly calibrated. On POPE-adversarial with visual contrastive decoding (Leng et al., 2024), adding grounding improves AURC from 0.1030 to 0.0603 and certified coverage from 47.6% to 64.2% ($\Delta\text{AURC} = 0.043 \pm 0.007$).

5.5 Baselines and attribution

Detector alone. Table 8 answers Remark 5's objection directly. CCRC beats a detector-only selective predictor in three of four settings, sometimes by very large margins, because the detector's correctness is harder to *score* than the VLM's. Where the detector genuinely dominates the VLM in accuracy (AMBER: 95.6% vs. 88.6%) it also wins standalone, which is a second reason CCRC is the wrong tool in that regime.

A published scorer, and where our margin comes from. Table 9 places the CLIP-similarity scoring function of Li et al. (2025) in our pipeline and adds our components one at a time. The aggregate margin is +67 points, and reporting only that number would misattribute it: +62.6 of the 67 comes from the score, and +4.4 from certified correction. We also note the comparison understates Li et al. (2025), whose method was designed for free-form caption claims rather than yes/no probes; we transplant their scorer, not their system. Figure 5 shows the corresponding risk–coverage curves.

The closest concurrent method. Xu et al. (2026) rescue borderline claims by acquiring zoomed views and re-scoring the *original* assertion. Holding the base score fixed and adding one mechanism per arm (Table 10), the two mechanisms are complementary: composing them beats every single-mechanism arm at

Table 9: Attribution on POPE ($\alpha = 0.10$). Most of the margin over a published scorer comes from the score, not from our procedure.

Method	Coverage	Δ
CLIP-similarity scorer (Li et al., 2025), filtering	5.6%	—
+ VLM confidence	41.5%	+35.9
+ detection grounding	68.2%	+26.7
+ certified correction (CCRC)	**72.6%**	+4.4

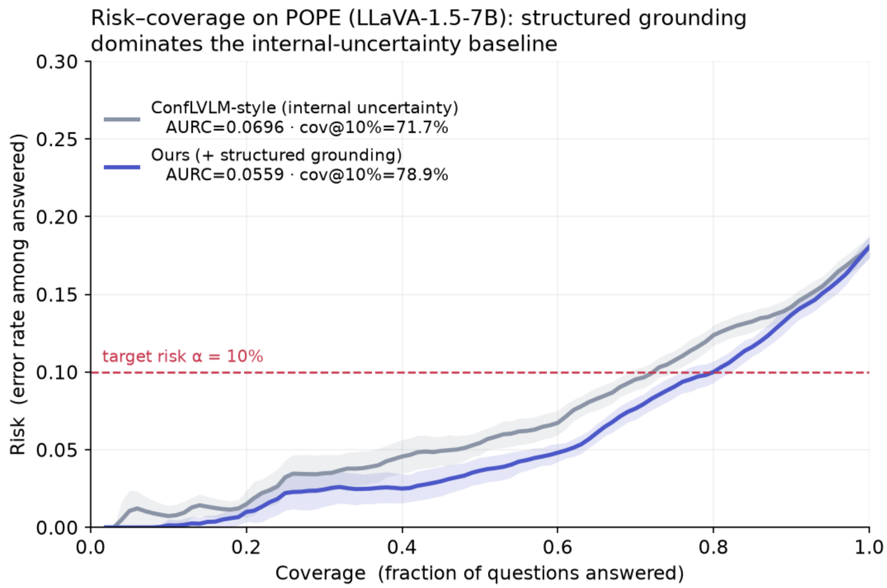

Figure 5: Risk–coverage curves on POPE. The grounded score (blue) lies below the internal-uncertainty baseline at every coverage level, i.e. lower error for the same coverage and more coverage for the same error target. Shaded bands are one standard deviation over 20 splits; the dashed line marks $\alpha = 0.10$.

both risk levels. Re-reading the image and replacing the answer rescue different claims. We do not claim to beat Xu et al. (2026): our reproduction uses three coarse crops and under-delivers relative to their published gain at $\alpha = 0.10$, so the defensible conclusion is composition, not superiority.

6 A sequential failure mode

Correction is more attractive in free-form generation than in yes/no answering: replacing a hallucinated word preserves a caption that deletion would mutilate. The extension looks mechanical — decompose the caption into claims, repair the unsupported ones, continue decoding — and two obstacles are already known. Editing a prefix makes later tokens off-policy, which is feedback covariate shift rather than exogenous shift (Fannjiang et al., 2022), so weighted conformal methods (Tibshirani et al., 2019) do not apply. And per-claim guarantees say nothing about a whole sequence, whose length is itself data-dependent.

Both could in principle be handled. Prefix-conditional (Mondrian) p-values make each decision valid given the realised prefix, so how we arrived there — including by our own edits — is absorbed into the conditioning; e-processes with Ville's inequality give sequence-level anytime validity, and are robust to adaptive intervention because our corrections are predictable with respect to the decoding filtration (Waudby-Smith & Ramdas, 2024; Ramdas et al., 2023). The residual difficulty is that the calibration pool drifts, since the deployed policy writes different text. The natural fix is on-policy calibration by iteration to a fixed point, and its

Table 10: One common base score, one mechanism added per arm (POPE, LLaVA-1.5, $n = 394$ grounded items). All arms valid. Composition beats every single mechanism.

Arm	cov@0.10	cov@0.15
Base filter (confidence only)	5.5%	16.5%
+ evidence acquisition (Xu et al., 2026)	5.5%	24.1%
+ detection-grounded score (ours)	32.1%	63.2%
+ certified correction (ours)	34.5%	71.8%
Composed	**35.8%**	**74.2%**

Table 11: Paired matched-prefix test of monotonicity ($n = 121$ cases). Repair significantly increases hallucination in the continuation.

Downstream measure (continuation only)	Original prefix	Repaired prefix
Hallucinated mentions (mean)	1.207	**1.413**
Hallucinations per 100 words	3.61	4.30
Faithful mentions (mean)	1.579	1.752
Continuation length (words)	33.4	32.9
Paired difference $+0.207 \pm 0.091$; $t = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$		
Cases better / worse / tie: 25 / 45 / 51		

convergence argument requires the correction map to be monotone. We measured whether it is, before attempting a proof.

Design. For each image we generate a caption greedily, locate the first hallucinated object mention at position t using AMBER's per-image present/absent object lists, and construct two prefixes *identical up to* t: one retaining the hallucinated word, one substituting the best-detected object the image genuinely contains. We then continue decoding the same number of tokens from each prefix and count hallucinated mentions in the continuations only. Matched prefixes make the contrast causal: the only difference between the two continuations is the intervention.

Result. Repair makes the continuation worse (Table 11, Figure 6). Downstream hallucinated mentions rise from 1.207 to 1.413, a paired difference of $+0.207 \pm 0.091$ ($n = 121$, $t = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$), with 45 cases worsening against 25 improving.

What this rules out. The correction map is not monotone, so the fixed-point argument has no basis: monotone convergence arguments of Knaster–Tarski type do not apply, and iterating calibration and deployment need not converge. We therefore do not claim a sequential guarantee of that form. What survives is a coupling bound, which is weaker but assumption-light.

For a sequence of claims $O = (o_1, \ldots, o_T)$ let $\mathcal{R}(O) = \mathbb{E}[\,\#\{t : o_t \text{ false}\}\,]$ be the expected count of false claims, the quantity a CHAIR-style loss charges for. Two regimes must be separated, and the distinction is exactly what §6 measures.

Theorem 2 (Coupling bound for corrected outputs)**.** *Let O denote the original output and O' the output after a repair rule is applied to a region R, and suppose the repair is* non-interfering*: conditional on not being repaired, every claim's truth value is unchanged by repairs elsewhere. Then*

$$\mathcal{R}(O') \;\le\; \mathcal{R}(O) \;+\; \Pr[X \in R] \cdot \Pr[\textit{repair wrong} \mid X \in R].$$

Without non-interference the bound acquires a third term,

$$\mathcal{R}(O') \;\le\; \mathcal{R}(O) \;+\; \Pr[X \in R] \cdot \Pr[\textit{repair wrong} \mid X \in R] \;+\; \mathcal{I}, \qquad \mathcal{I} \;:=\; \mathbb{E}\big[\Delta_{R^c}\big], \tag{4}$$

where Δ_{R^c} is the change in the number of false claims outside R induced by repairing inside it.

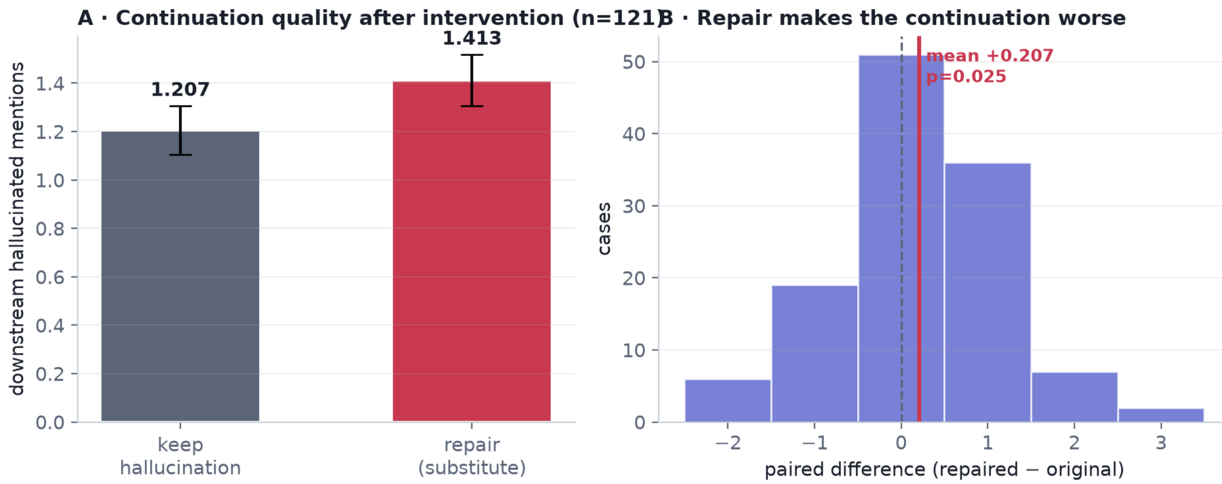

Figure 6: (A) Mean hallucinated mentions in the continuation after keeping versus repairing the flagged claim; error bars are standard errors. (B) Distribution of the paired difference; mass to the right of zero indicates repair worsening the continuation. The effect is modest in size but statistically clear, and its sign is what matters for the theory.

Proof. Decompose by whether a claim is repaired. Under non-interference the claims in R^c retain their truth values, contributing at most $\mathcal{R}(O)$; on R the emitted claim is wrong only if the repair is wrong, contributing at most $\Pr[X \in R] \Pr[\text{repair wrong} \mid X \in R]$. Summing gives the first bound. In general the R^c contribution is $\mathcal{R}(O) + \mathbb{E}[\Delta_{R^c}]$ by definition of Δ_{R^c}, which gives (4). $\square$

Remark 6 (The interference term is not negligible here). Non-interference is the assumption §6 falsifies: we measure $\widehat{\mathcal{I}} = +0.207 \pm 0.091$ additional false claims per repair ($n = 121$, $p = 0.025$), so for a single repair the interference term is roughly the same size as the direct repair-error term and cannot be dropped. Any procedure certifying corrected sequences must estimate $\mathcal{I}$ rather than assume it away, and $\mathcal{I}$ depends on the repair rule — which is why a context-aware rule is the natural next step rather than a tighter bound.

Remark 7 (What the corollary does and does not give). Controlling $\mathcal{R}(O) \le \alpha_1$ and the repair term by α_2 at confidence $1 - \delta/2$ each yields $\mathcal{R}(O') \le \alpha_1 + \alpha_2 + \mathcal{I}$ at confidence $1 - \delta$. We flag a limitation of this composition rather than hiding it: certifying $\mathcal{R}(O)$ for the *uncorrected* generation is precisely the object the abstention floor makes expensive, so the corollary is useful only relative to an already-selective base policy (for which $\mathcal{R}$ is conditioned on emission), not to raw generation. Making the sequential guarantee self-contained — certifying the corrected sequence without first certifying the uncorrected one — remains open.

What this implies for algorithm design. Beyond closing off a proof strategy, the measurement identifies a cost that a myopic procedure does not charge for. Greedy per-claim repair optimises the risk of the claim in front of it while imposing roughly 0.21 additional hallucinated claims on the continuation. A correct sequential procedure must therefore price the *continuation*, not the claim — for instance by searching over rewrite trajectories under an accumulated risk ledger rather than deciding each claim independently. We regard this as the main open problem the paper leaves, and note that it is a genuinely different statistical object: because repairing claim i changes which claims $i + 1, \ldots$ exist, the item set is endogenous, and selecting among trajectories introduces a selective-inference problem that per-item conformal methods do not face.

A competing explanation, and what survives it. Table 11 contains a row that complicates the headline. Repair raises faithful mentions by +0.174 at the same time as it raises hallucinated mentions by +0.207, in continuations that are slightly *shorter* (33.4 → 32.9 words). Repaired prefixes therefore produce denser object talk, and the obvious alternative reading is that correction makes the continuation

Table 12: Combiner ablation on POPE ($\alpha = 0.10$, 30 repetitions). Under data reuse, high-capacity combiners *break* the guarantee; under a three-way split they are valid and tied. Logistic regression is chosen deliberately.

	Naive: fit = calibrate		Three-way split	
Combiner	cov@10%	test risk	cov@10%	test risk
Logistic regression	74.8%	8.8%	72.5%	8.4%
Gradient boosting	91.3%	**13.3%** (violated)	72.5%	8.3%
Random forest	88.5%	**12.1%** (violated)	75.7%	8.5%
MLP $(64, 32)$	≈28%	—	≈28%	—

more descriptive rather than less faithful. We tested this directly with two normalisations. Per word, hallucination rises significantly (+0.0075 per word, $p = 0.011$). Per *mention*, it does not move at all: the hallucinated share of mentions goes $0.433 \to 0.447$ pooled, and the paired difference is -0.004 ($p = 0.884$, $n = 111$ pairs with mentions in both arms). The competing explanation is therefore substantially correct as a description of *why* the count rises.

It does not, however, rescue the monotonicity argument, and the reason is that the quantity being certified is a count and not a ratio. A CHAIR-style risk charges for each hallucinated mention emitted; a procedure that repairs one claim and emits two additional false ones has increased the risk of its own output regardless of how many true claims came with them. What the normalised results do establish is the mechanism — repair perturbs the generation towards denser description, and hallucination scales with density — and that a density-controlled repair rule might avoid the cost. We report the share result because a reader is entitled to it, and because it makes the negative result narrower than the raw count suggests.

Scope of the negative result. Our repair substitutes the globally best-detected object, which can be contextually inappropriate mid-sentence and may push the prefix off-policy for reasons unrelated to faithfulness. The supported claim is that *context-blind* substitution carries a downstream cost in absolute and per-word terms, not that all correction must, and not that it degrades the faithful/hallucinated *balance*. Constraining repairs to the model's own top-k continuations — which the one-shot procedure of §4 already implies — is the natural remedy and remains to be tested.

7 Ablations

7.1 The score combiner, and a validity trap

Channel 1 is a learned function of a handful of scalars. It is natural to ask whether a more expressive combiner helps. It does not, and it can silently invalidate the guarantee.

Table 12 reports two protocols. Under the naive protocol, where the combiner is fitted on the same fold used to calibrate τ, gradient boosting and random forests appear to certify far more coverage — 91.3% and 88.5% — but their *realised* test risk is 13.3% and 12.1% against a 10% target. The mechanism is straightforward: a high-capacity model overfits the calibration scores, so the empirical error on the calibration fold understates the true error, and the threshold is set too permissively. Under a three-way split all combiners are statistically tied and all are valid, while a small multilayer perceptron collapses ($\approx$ 28% coverage, high variance) from overfitting four features.

We draw two conclusions. Practically, the score must be fitted on a fold disjoint from calibration — a discipline that costs coverage but is necessary for any scorer whose capacity is not negligible. Conceptually, capacity is not the bottleneck: the features are already informative scalars, and what limits efficiency is the quality of the grounding signal, not the flexibility of the combiner.

Table 13: Risk-bound ablation (POPE, $\alpha = 0.10$, three-way split). Exact binomial dominates for a binary loss; the others are valid but conservative.

Upper confidence bound	cov@10%	realised risk
Clopper–Pearson (exact)	**72.5%**	8.4%
Empirical Bernstein	54.9%	4.6%
Hoeffding	38.3%	3.2%

Figure 7: (A) Risk–coverage on POPE for the score variants. (B) Capability comparison with recent methods. Mitigation approaches change the base model but provide no guarantee; existing conformal approaches guarantee but only remove or abstain. Our layer composes on top of mitigation decoders (§5).

7.2 The risk bound

The emitted-set error is a binomial proportion, so an exact interval should dominate general-purpose concentration inequalities, and it does: at $\alpha = 0.10$ Clopper–Pearson certifies 72.5% coverage, empirical Bernstein 54.9%, and Hoeffding 38.3% (Table 13). All three are valid; the latter two are simply conservative, realising 4.6% and 3.2% risk against a 10% budget. This is consistent with the hierarchy established by Kotte (2026a) and indicates that for binary losses the choice of bound is worth more than several points of coverage. Variance-adaptive or betting-style bounds (Waudby-Smith & Ramdas, 2024) become preferable for graded losses, which is the natural setting for claim-level factuality scores.

7.3 Positioning summary

Figure 7 summarises where CCRC sits. Panel A gives risk–coverage curves; panel B contrasts capabilities across recent methods. Mitigation methods (Huang et al., 2024; Jiang et al., 2025; Wu et al., 2025) operate only at full coverage and offer no guarantee; conformal methods (Li et al., 2025; Lee et al., 2025) guarantee but only delete or abstain; Xu et al. (2026) re-scores without changing the answer. CCRC is, to our knowledge, the first to certify an output that differs from the model's — while requiring, as §4.4 shows, an independent channel to do so in practice.

8 Discussion

What the results support. Permitting a selective predictor to emit a substituted answer is a real degree of freedom: it evades the premise of Proposition 1 and, when the preconditions hold, converts a small amount of high-precision external evidence into several times as much certified coverage. The guarantee is unaffected by the substitution (Theorem 1), because risk control cares about error indicators rather than about provenance.

What they do not. The gains are modest and conditional. The mechanism requires headroom (Proposition 4), a strict gate (Table 3), and an independent channel (§4.4); each is a real constraint, and one of our five benchmarks fails the first. Most of our margin over a published baseline comes from the correctness score rather than from the algorithm (Table 9), and the closest concurrent method is complementary rather than dominated (Table 10). The one-shot theory is a correct but standard application of fixed-sequence testing with exact bounds; the interesting theory is sequential, and §6 shows the most natural route to it is closed.

Limitations. Channel 2 is an object detector, so the evaluated method addresses existence hallucination; on attribute-, counting- and reasoning-heavy benchmarks it degenerates to filtering, which remains valid but gains nothing. The independence of channel 2 is an assumption, and a detector sharing the VLM's blind spots would weaken certification without invalidating it. The repair gate q is fixed a priori rather than learned, and a data-driven choice would require its own share of the confidence budget. Exchangeability is assumed throughout; under deployment shift the stated level need not hold, and adaptive conformal methods (Gibbs & Candès, 2021) would be required. Finally, our sequential experiment uses one backbone and a context-blind repair rule, and its negative conclusion should be read with that scope.

Outlook. The open problem we consider most interesting is the one the negative result exposes. Because repairing a claim changes which subsequent claims exist, a sequential procedure faces an endogenous item set and must select among rewrite trajectories rather than filter a fixed list. Selecting the best trajectory and then reporting its certificate is a selective-inference error, so a correct procedure needs a certificate that is simultaneous over the search tree. Building that, and pricing the downstream cost we measured, is the natural continuation of this work.

9 Conclusion

Selective prediction for vision–language models is expensive because of a bound whose premise — that the emitted answer is the model's own — is an assumption about the action space rather than about the data. Relaxing it works, but not freely: certified correction requires an independent evidence channel, a strict repair gate, and base risk comfortably above the target, and it delivers its gains through a dilution effect that gives small repairs disproportionate leverage. Applied myopically in generation, the same idea makes things measurably worse. We have tried to report the conditions and the failures as carefully as the improvements, in the belief that for methods whose selling point is a guarantee, the boundaries of that guarantee are the substance of the contribution.

Broader Impact Statement

Attaching guarantees to model outputs can improve safety, but a guarantee is only as meaningful as its assumptions. Our procedure controls the error rate among emitted answers under exchangeability and requires a genuinely independent evidence channel; if either fails — under deployment shift, or with a detector sharing the model's blind spots — the stated level may not hold. Emitting a *corrected* answer also changes the failure mode presented to a user, replacing an abstention that invites scrutiny with a confident substitution that may not. For this reason we regard the checkable precondition and the sequential negative result as safety-relevant components of the contribution rather than limitations to be minimised, and we recommend that any deployment verify the precondition and monitor channel independence rather than treating the nominal α as a guarantee that transfers unconditionally.

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A Additional results

A.1 Synthetic validation of the three pillars

Before any real-data experiment we validated the procedure on synthetic data where ground truth is controllable, confirming (i) that the certified risk is respected, (ii) that a better-separated score buys coverage at fixed risk, and (iii) that a single global threshold can satisfy a marginal guarantee while violating it on a minority subgroup (25.3% realised risk against a 10% target), which group-conditional (Mondrian) calibration repairs (9.8% and 9.7% on the two groups). The last point motivates treating conditional coverage as a separate requirement from marginal coverage; we report it on synthetic data only, as our real-data samples are not category-balanced.

A.2 Multi-α table

Table 14 gives certified coverage at four risk levels, the numerical version of Figure 4.

Table 14: Certified coverage at four risk targets (POPE, LLaVA-1.5, 20 splits).

Score	$\alpha = 0.05$	0.10	0.15	0.20
Confidence only	30.0%	65.2%	84.6%	99.1%
+ detection grounding	**52.5%**	**73.6%**	**89.7%**	**99.6%**

A.3 Reproducibility

All extracted per-item signals (model probabilities, self-consistency counts, detector scores, labels) are released as JSON, so every table and figure in this paper can be regenerated on CPU without access to a GPU. The GPU scripts that produced those signals are released as well, together with the failed variants discussed in §4.5 and §4.4, which we retain because the failures are part of the argument.